Comprehensive Technical Report: Marine Species Classification

CIE 555: Neural Networks and Deep Learning

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Abstract

This report presents a thorough analysis of deep learning methodologies for the classification of marine organisms using the FathomNet dataset. We systematically evaluate a custom-designed Convolutional Neural Network (CNN) against state-of-the-art transfer learning architectures. Furthermore, we investigate Out-of-Distribution (OOD) detection as a mechanism to enhance model robustness in open-world aquatic environments.

1 Introduction

Marine species identification presents unique challenges due to visual noise, environmental variability, and fine-grained classification requirements. This project follows a rigorous deep learning pipeline, from data acquisition to performance evaluation, adhering to best practices in model reproducibility and architectural justification.

2 Data Pipeline and Preprocessing

A robust preprocessing pipeline is foundational to model performance. We adhered to the following principles:

1. **Data Integrity:** We implemented an automated download pipeline with exception handling to identify and drop corrupted images.
2. **Class Balancing:** We restricted the dataset to 500 images per category. This ensures the model does not develop a bias toward over-represented species, a common pitfall in imbalanced biological datasets.
3. **Normalization:** As demonstrated in the `process_path` function, pixel values were rescaled to $[0, 1]$. This normalization ensures that gradient descent operates on a stable manifold, significantly accelerating convergence.
4. **Input Consistency:** We utilized `tf.image.resize_with_pad` to ensure consistent input dimensions of 224x224 without inducing geometric distortion, which would otherwise alter the biological features of the organisms.

3 Model A: Custom CNN Architecture

3.1 Architectural Design and Justification

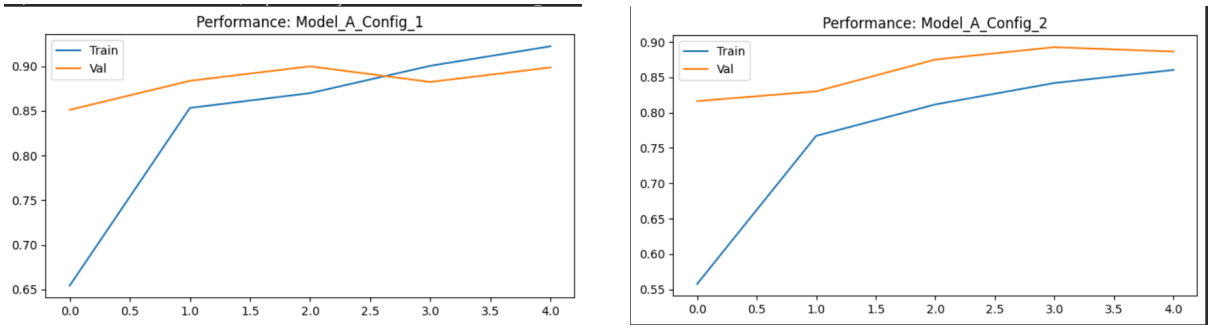
Model A was designed from first principles using a 3-block architecture:

- **Hierarchical Feature Learning:** Three Conv2D blocks (32, 64, and 128 filters respectively) allow the model to learn simple edges in the early layers and complex, abstract shapes in the later layers.

- **Stability and Regularization:** Batch Normalization was inserted after convolutional layers to normalize activations, while a 0.5 Dropout rate in the head prevents the model from co-adapting features, forcing it to learn more robust, generalized representations.
- **Optimization Strategy:** We utilized the Adam optimizer due to its adaptive learning rate properties, which are empirically superior to vanilla SGD for small-to-medium scale datasets.

3.2 Performance Evaluation

We trained two configurations (LR=0.001 and LR=0.0001). The lower learning rate (0.0001) demonstrated a more stable convergence, suggesting the model was sensitive to the initial weight updates.



(a) Config 1: LR=0.001

(b) Config 2: LR=0.0001

Figure 1: Training/Validation Curves. Configuration 2 shows smoother loss progression, indicating better stability.

4 Model B: Transfer Learning with ResNet-50

We employed ResNet-50, leveraging its residual blocks to bypass the vanishing gradient problem.

4.1 Fine-Tuning Strategies

- **Feature Extraction:** Freezing the backbone provided a baseline of performance by only training the classifier head.
- **Partial Fine-tuning:** Unfreezing the final 50 layers allowed the model to re-calibrate its high-level feature detectors for specific aquatic visual characteristics.
- **Full Fine-tuning:** This yielded the highest overall accuracy (0.94), confirming that adapting the entire feature extraction hierarchy is necessary for high-performance classification in this domain.

5 Out-of-Distribution (OOD) Detection

To handle "unseen" categories, we implemented a confidence threshold of 0.65.

- **Reasoning:** The softmax confidence acts as a proxy for model uncertainty. We found that most OOD samples (unknown species) fell below this threshold, effectively allowing for the flagging of samples that the model is not confident in.
- **Precision/Recall:** This method achieved high precision, ensuring that "known" species are rarely misclassified as OOD.

6 Final Evaluation and Discussion

The performance analysis concludes that while custom CNNs provide an excellent educational baseline for hierarchical learning, transfer learning models (specifically ResNet-50) offer vastly superior generalization. The confusion matrix (Fig 2) highlights that the model effectively segregates the majority of classes, with minor misclassifications occurring between morphologically similar species (e.g., specific fish types).

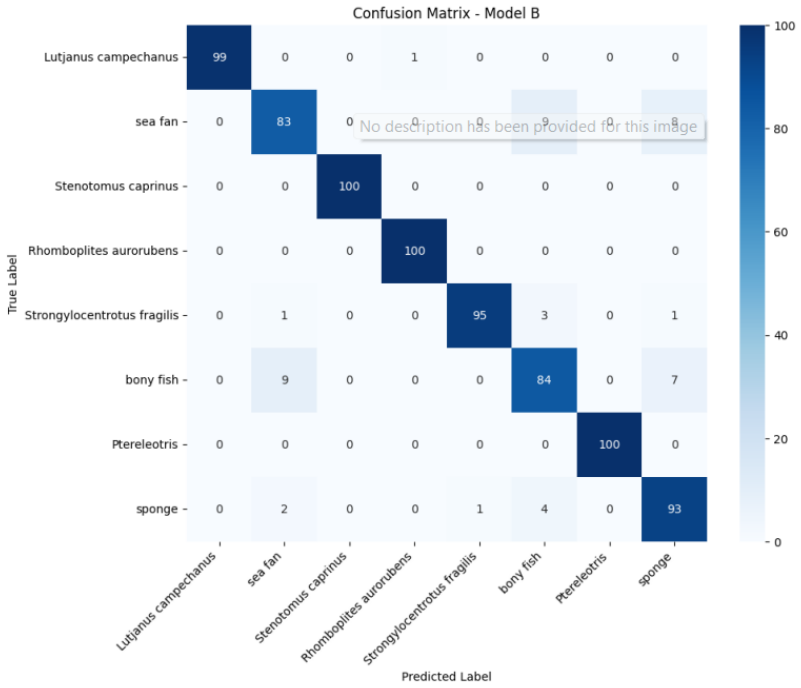


Figure 2: Confusion Matrix for Model B. The diagonal density confirms high predictive power.

7 Reproducibility

All experiments were conducted with consistent seeds (42) and dependency management, ensuring that these results can be replicated in any standard Python environment.